

### Set 3: Employee Performance Evaluation

```
library(ggplot2)
library(plotly)
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c(
    "Sales",
    "HR",
    "Marketing",
    "Sales",
    "HR"
  ),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)
print(employee_data)
line_chart <- ggplot(
  employee_data,
  aes(
    x = Years_of_Service,
    y = Performance_Score
  )
) +

  geom_line(
    linewidth = 1
  ) +

  geom_point(
    size = 3
  ) +
```

```

labs(
  title = "Employee Performance Trend Over Years of Service",
  x = "Years of Service",
  y = "Performance Score"
) +

theme_minimal()

print(line_chart)

department_count <- table(
  employee_data$Department
)

department_data <- data.frame(
  Department = names(department_count),
  Employees = as.numeric(department_count)
)

print(department_data)

bar_chart <- ggplot(
  department_data,
  aes(
    x = Department,
    y = Employees
  )
) +

geom_bar(
  stat = "identity"
) +

labs(

```

```
title = "Distribution of Employees by Department",  
x = "Department",  
y = "Number of Employees"  
) +
```

```
theme_minimal()
```

```
print(bar_chart)
```

```
scatter_plot <- ggplot(  
  employee_data,  
  aes(  
    x = Years_of_Service,  
    y = Performance_Score  
  )  
) +
```

```
geom_point(  
  size = 3  
) +
```

```
geom_smooth(  
  method = "lm",  
  se = FALSE  
) +
```

```
labs(  
  title = "Years of Service vs Performance Score",  
  x = "Years of Service",  
  y = "Performance Score"  
) +  
theme_minimal()
```

```

print(scatter_plot)

correlation <- cor(
  employee_data$Years_of_Service,
  employee_data$Performance_Score
)

print(
  paste(
    "Correlation between Years of Service and Performance Score:",
    round(correlation, 2)
  )
)

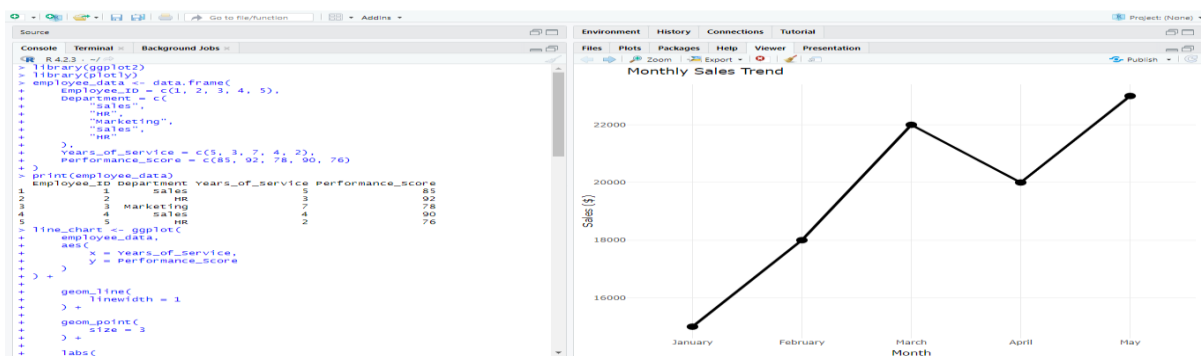
interactive_line_chart <- ggplotly(
  line_chart
)

interactive_bar_chart <- ggplotly(
  bar_chart
)

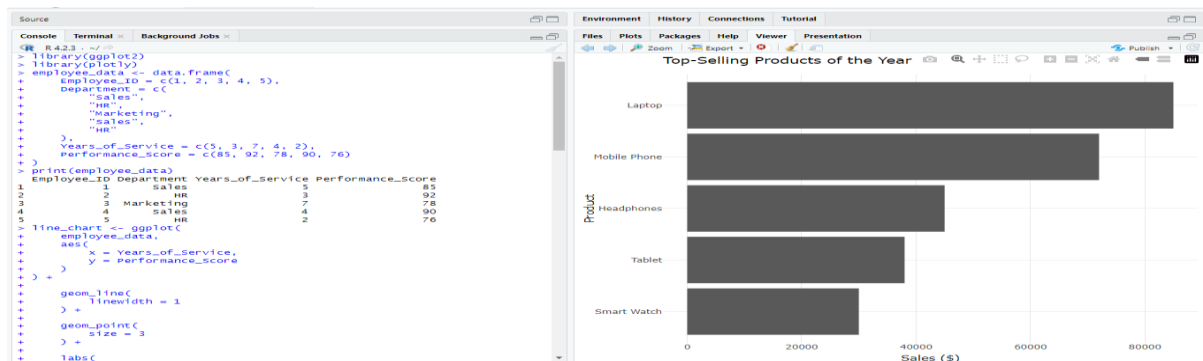
interactive_line_chart
interactive_bar_chart

```

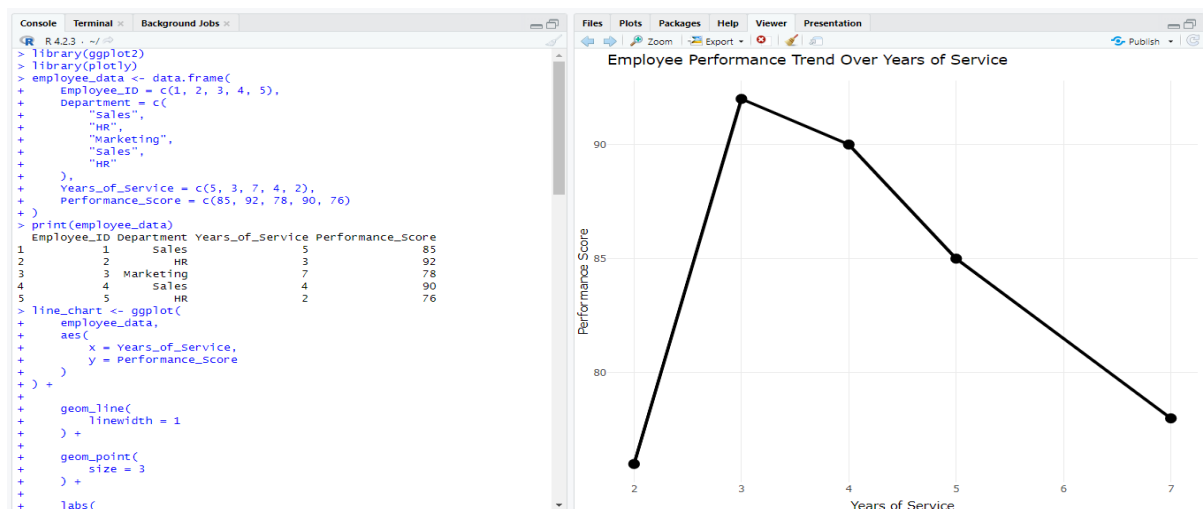
**1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.**



**2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.**



**3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.**



**4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.**

